

Invariant-Structured Online Control for Safe Autonomy

(With a Phase-First Treatment of Two-Way Doppler Metrology)

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Abstract

Constructive autonomy builds the right optimization problem for the current scene and adapts it online as state, uncertainty and intent evolve. An invariant-structured control stack is presented in which three orthogonal invariants—a continuous log-deviation ξ , a discrete structural indicator S , and a non-equilibrium knob `sct`—serve as constraint composers, boundary encoders and symmetry-aware regularizers inside model predictive control and control-barrier safety filters. The formulation includes formal safety margins for forward invariance, solver-adaptation schedules keyed to `sct`, and a provenance-to-knobs pipeline that generalizes across domains. A full *reductions* section maps invariants to canonical optimization forms and to invariant-set guarantees. We also introduce a *phase-first two-way Doppler* module: sensing is framed as measuring a *phase rate on a single clock*, which yields an operational, parallel EM/acoustics treatment and exposes the convective term for flow. The result enables coherent, safety-preserving re-solves under perturbations and supports rapid online adaptation without sacrificing feasibility.

Context and summary. Autonomy increasingly depends on invariants to keep plans safe and consistent under model error and disturbance. The approach below injects continuous and discrete invariants into both the optimization *and* the solver, and aligns the perception loop via phase-rate Doppler metrology that naturally integrates with range/range-rate estimates for closed-loop control.

Background and related work

Invariant sets and safety in planning and control

Planning methods routinely attach a feedback controller to each candidate path and use the associated controlled-invariant set as a safety certificate. Model predictive control (MPC) schemes constrain the terminal state to a control-invariant safe set to guarantee recursive feasibility. Exact invariant sets are difficult for nonlinear systems, so practical designs learn or approximate these sets online. Receding-constraint MPC propagates a learned safe set backward over the horizon. Tube/robust MPC maintains an invariant tube around the nominal trajectory to bound errors; contraction-based MPC shrinks robust control-invariant tubes by optimizing contraction metrics, expanding feasible operating regions.

Constraint and objective adaptation in real time

Modern planners adapt objectives and constraints online; invariant sets structure these adaptations. Receding-constraint MPC treats the previously computed invariant set as a moving boundary. Robust formulations adjust terminal sets or disturbance tubes on the fly, tightening constraints when conditions worsen. Invariants thus act as live boundaries that shift as the scene evolves, preserving feasibility and safety.

Symmetry and invariance in learning-based control

Symmetries reduce model and policy complexity. Moving-frame constructions yield models invariant to specified transformations; neural architectures encode equivariance so actions transform coherently with state. Such symmetry-aware designs operate as structural regularizers that limit exploration to invariant subspaces.

Physics-based invariants and safety monitoring

Physical invariants (e.g., balance laws or linearized relationships) support anomaly detection and consistency checks complementary to planning. Learned invariants provide real-time safety bounds and can be embedded as constraints in decision modules.

Alignment with invariants (ξ, S)

Two orthogonal diagnostics guide the stack: a continuous log-deviation ξ and a discrete structural parity S . ξ parallels smooth Lyapunov/RCI levels and continuous symmetry actions; S behaves like a guard in hybrid models, toggling discrete boundaries. Together they form multi-scale invariants: ξ for smooth deviation control, S for mode activation.

Phase-First Doppler Metrology: EM–Acoustics with a Single-Clock Phase Rate

Unifying stance. We measure Doppler as a *phase rate on a single clock*. Every link carries a scalar phase ϕ ; the station observes the derivative of a *phase difference*. For EM, phase propagates at invariant speed c with Lorentz kinematics. For acoustics, phase rides a moving medium with speed c_s in the medium frame; in the lab frame, the phase rate includes a convective term from the flow $\mathbf{u}(\mathbf{x}, t)$. A coherent two-way echo becomes an operational phase-accounting loop in which range and range-rate map directly into a single-clock phase-rate measurement.

Core definitions (operational)

Definition 1 (Phase along a ray). *For a narrowband wave along path γ ,*

$$\phi(t) = \int_{\gamma} \mathbf{k} \cdot d\mathbf{x} - \int \omega dt, \quad f_r(t) = \frac{1}{2\pi} \frac{d\phi_r}{dt}. \quad (1)$$

Definition 2 (Two-way phase difference (coherent turnaround M)). *Let $\phi_{\downarrow}(t)$ be the received downlink phase and $\phi_{\uparrow}(t)$ the (time-shifted) uplink phase. Define*

$$\Delta\phi(t) = \phi_{\downarrow}(t) - M\phi_{\uparrow}(t - \tau), \quad y(t) \equiv \frac{f_r(t) - Mf_0}{Mf_0} = \frac{1}{2\pi Mf_0} \frac{d}{dt} \Delta\phi(t), \quad (2)$$

where f_0 is the station clock and τ is the round-trip time.

Remark 1 (Characteristic (phase) speed). *We write v_{ϕ} for the characteristic speed: for EM, $v_{\phi} = c$; for acoustics measured in the medium frame, $v_{\phi} = c_s$.*

Two-way echo as a phase machine

E1. Transmit carrier with known phase $\phi_{\uparrow}(t)$ and clock f_0 .

E2. Remote target returns a coherent echo at ratio M (or a locked transponder).

E3. Mix the downlink with local Mf_0 to form $\Delta\phi(t)$.

E4. The fractional Doppler $y(t)$ is the *time derivative of this phase difference on the same clock*, as in (2).

No remote clock is needed; the world modulates the out-and-back phase.

EM vs. acoustics: identical bookkeeping, different kinematics

Electromagnetics (vacuum; invariant c). For a plane wave $\phi = \mathbf{k} \cdot \mathbf{x} - \omega t$ with $\omega = c\|\mathbf{k}\|$, the two-way small-velocity fractional Doppler for radial speed $v_r = \dot{R}$ is

$$y_{\text{EM}}(t) \approx -\frac{2v_r(t)}{c} + (\text{accel} / \text{gravitational} / \text{relativistic corrections}). \quad (3)$$

Interpretation: the light-time $\tau = 2R/c$ slides as R changes, and $\dot{\Delta\phi}$ tracks that slide.

Acoustics (preferred frame = medium; flow $\mathbf{u}$). Lab-frame dispersion is Galilean:

$$\omega = c_s\|\mathbf{k}\| + \mathbf{k} \cdot \mathbf{u}, \quad (4)$$

so $\mathbf{k} \cdot \mathbf{u}$ is the *convective phase-rate* term. For one-way source/receiver speeds $\mathbf{v}_s, \mathbf{v}_r$ and propagation direction $\hat{\mathbf{n}}$,

$$f_r = f_s \frac{c_s + \hat{\mathbf{n}} \cdot (\mathbf{u} - \mathbf{v}_r)}{c_s + \hat{\mathbf{n}} \cdot (\mathbf{u} - \mathbf{v}_s)}. \quad (5)$$

86 For a coherent two-way echo (small velocities, same ray out/back),

$$87 \quad y_{ac}(t) \approx -\frac{2}{c_s} \hat{\mathbf{n}} \cdot (\mathbf{v}_r(t) - \mathbf{u}(t)) + (\text{range-accel} / \text{inhomogeneity corrections}). \quad (6)$$

88 Read it as “replace c by c_s relative to the medium and subtract the flow along the ray.”

89 Practical notes and figure hooks

- 90 • **Unifying one-liner:** In both EM and acoustics we observe a *phase rate* on the ground clock;
 91 kinematics only enter by how they map geometry and flow into $\Delta\dot{\phi}$.
- 92 • **Figure ideas:** (i) Two-way phase loop (single-clock mixing diagram). (ii) Parallel EM/acoustics
 93 “same algebra, different v_ϕ ”. (iii) Convective baseline shift from $\mathbf{k} \cdot \mathbf{u}$.

94 Results

95 System model, safety set, and invariants

96 Consider control-affine dynamics $x_{t+1} = f(x_t) + g(x_t)u_t + w_t$ with safety set $\mathcal{C} = \{x : h(x) \geq 0\}$.
 97 Over a rolling window W , a local LTI surrogate $H(s) = k \prod(s - z_i) / \prod(s - p_j)$ is identified by the
 98 provenance pipeline (§). Define

$$99 \quad \Lambda_S = k, \quad \Lambda_G = \text{GM}_W[k], \quad \xi = \ln(\Lambda_S / \Lambda_G), \quad S = \mathbf{1}[n_z \neq n_p], \quad \varepsilon = \text{median}(|\xi|) + \alpha \text{Var}(\mathbf{sct}) + \beta \mathbf{1}[S = 1]. \quad (7)$$

100 Here ξ is a continuous deviation, S a discrete parity, and ε is a composite conditioning index with
 101 weights $\alpha, \beta > 0$.

102 Invariant-structured MPC

103 At decision time t , solve

$$104 \quad \min_{u_{0:N-1}} \sum_{k=0}^{N-1} \ell(x_k, u_k; \xi_t) + \lambda_{\text{sym}}(\xi_t) \mathcal{R}_{\text{sym}}(x_{0:N-1}, u_{0:N-1}) \quad (8)$$

$$105 \quad \text{s.t. } x_{k+1} = f(x_k) + g(x_k)u_k, \quad (x_k, u_k) \in \mathcal{Z}(\xi_t, S_t), \quad x_N \in \mathcal{X}_{\text{inv}}(\varepsilon_t),$$

$$106 \quad p^\top f(x_t) + p^\top g(x_t)a + q \geq (1 - \eta)h(x_t) - \delta(\varepsilon_t). \quad (9)$$

107 $\mathcal{Z}$ composes environment and comfort constraints by (ξ, S) ; $\mathcal{X}_{\text{inv}}$ shrinks with ε ; $\mathcal{R}_{\text{sym}}$ penalizes
 108 deviations from task symmetries (e.g., $SE(2)$ equivariance).

109 Safety filter (CBF-QP)

110 Given a nominal policy $\pi(x)$,

$$\begin{aligned}
 111 \quad a_t^* &= \arg \min_{a \in \mathcal{U}} \|a - \pi(x_t)\|^2 \\
 112 \quad \text{s.t. } & p^\top f(x_t) + p^\top g(x_t)a + q \geq (1 - \eta)h(x_t) - \delta(\varepsilon_t).
 \end{aligned} \tag{10}$$

113 Theory: recursive feasibility and forward invariance

114 **Assumption 1** (Lipschitzness and bounded errors). *Over $\mathcal{X} \times \mathcal{U}$, f, g are Lipschitz. Prediction/encoding errors satisfy $\|\Delta f\| \leq \epsilon_f$, $\|\Delta g\| \leq \epsilon_g$, encoder error $\leq \epsilon_z$.*

116 **Proposition 1** (Invariant-shaped robust margin). *If*

$$117 \quad \delta(\varepsilon) \geq \epsilon_f \|p\|_1 + \epsilon_g \|p\|_1 \max_{a \in \mathcal{U}} \|a\|_1 + \epsilon_z (L_f + L_{f_\theta}) + \epsilon_z \max_{a \in \mathcal{U}} \|a\|_1 (L_g + L_{g_\theta}), \tag{11}$$

118 *then any a_t^* from (10) guarantees $h(x_{t+1}) \geq (1 - \eta)h(x_t)$ and forward invariance of $\mathcal{C}$.*

119 *Sketch.* Bound the one-step barrier deficit by Lipschitz constants and errors $(\epsilon_f, \epsilon_g, \epsilon_z)$; choose $\delta(\varepsilon)$
 120 to dominate the worst-case deficit; apply the CBF recursion. $\square$

121 **Lemma 1** (Recursive feasibility under $\mathcal{X}_{\text{inv}}(\varepsilon)$). *Suppose $\mathcal{X}_{\text{inv}}(\varepsilon)$ is robust positively invariant for*
 122 *the closed-loop dynamics under the IS-MPC policy with margin $\delta(\varepsilon)$. If $x_N \in \mathcal{X}_{\text{inv}}(\varepsilon_t)$ at time t ,*
 123 *then the shifted plan at time $t+1$ remains feasible provided $\varepsilon_{t+1} \leq \bar{\varepsilon}$ for some schedule bound $\bar{\varepsilon}$.*

124 *Sketch.* Use invariance of $\mathcal{X}_{\text{inv}}$ and warm-started MPC feasibility arguments (shifted sequence plus
 125 terminal admissibility). The schedule bound ensures the terminal set does not shrink faster than
 126 the reachable set. $\square$

127 Reductions: from invariants to optimization structure

128 The invariants map to canonical components:

129 **R1. Cost scaling (continuous reduction).** ξ rescales quadratic penalties and comfort terms:

$$130 \quad \ell(x, u; \xi) = \omega(\xi) \ell_0(x, u) \text{ with } \omega \text{ monotone in } |\xi|.$$

131 **R2. Constraint gating (discrete reduction).** S selects boundary-condition encoders $\mathcal{Z}_0$ or $\mathcal{Z}_1$
 132 (e.g., lane-keep vs. lane-change guards).

133 **R3. Feasibility margin (conditioning reduction).** ε schedules $\delta(\varepsilon)$ and terminal-set size in
 134 $\mathcal{X}_{\text{inv}}(\varepsilon)$, ensuring recursive feasibility under uncertainty.

135 **R4. Solver schedule (computational reduction).** sct controls horizon and trust-region radii:
 136 $N(\varepsilon)$ and $\Delta u(\varepsilon)$ trade optimality for latency with guarantees preserved by (11).

137 **R5. Symmetry regularization (structure reduction).** $\lambda_{\text{sym}}(\xi)$ increases with $|\xi|$, emphasizing
 138 equivariance when conditions deteriorate to stabilize learning and optimization.

Algorithm and complexity

Algorithm 1 Invariant-Structured MPC (IS-MPC)

- 1: **Inputs:** state x_t , prior plan $(x, u)_{0:N-1}$
 - 2: Identify $H(s)$ on a rolling window W ; compute $\xi_t, S_t, \mathbf{sct}_t$ and ε_t
 - 3: Compose (8)–(9) conditioned on $(\xi_t, S_t, \varepsilon_t)$
 - 4: Set $N \leftarrow N(\varepsilon_t)$ and trust radius schedule; warm-start
 - 5: Solve MPC; if infeasible, increase $\delta(\varepsilon_t)$ and shrink trust radius; retry
 - 6: Apply u_t ; if using a nominal policy, project via (10)
 - 7: Shift horizon; update W
-

Per-tick costs: local ID $O(n^3)$; composing constraints $O(1)$; dense-QP per-iteration $O(Nm^3)$ with input dim m . Warm-starting maintains a small iteration count at control rates of 50–100 Hz.

Ablations and expected outcomes

- **No symmetry regularizer:** higher equivariance error; degraded generalization.
- **No composer:** more constraint violations under distribution shift.
- **Fixed δ :** either unnecessary conservatism or unsafe transients.
- **Fixed N :** worse latency/feasibility trade under fast perturbations.

Methods

Provenance pipeline: from data to invariants

The following steps produce $(\xi, S, \mathbf{sct})$ and generalize across domains:

- M1. Normalization and CCA:** per group normalization to the smallest-variance variable; logarithmic ratios; two-pass CCA to interlink three groups and obtain representative coordinates (E_1, E_2, P_1, S_1) .
- M2. Geometric reduction:** build triangles $\triangle E_1 E_2 S_1$ and $\triangle E_1 E_2 P_1$ with centroids (G_s, G_p) ; compute $d = \|G_p - G_s\|_2$ and triangle area K .
- M3. Representative partition:** compute sectional features $s_{1..4}$ from $[d, k, T, ER, SBR]$ to obtain robust, two-tier descriptors (size and identity).
- M4. Neural lab and S_c :** probe three units at three scales; aggregate 5×5 traces to score vectors; form the symmetric characteristic matrix $S_c = \frac{1}{2}(S_{AB}S_{BA}^\top + S_{BA}S_{AB}^\top)$.
- M5. Specific computational time:** measure $\mathbf{sct}$ by performing a fixed characteristic operation on S_c (vectorization and sorting) under a fixed environment.

M6. Elliptic/LTI stage: obtain (z, p, k) and define $H(s) = k \prod (s - z_i) / \prod (s - p_j)$.

M7. Invariants: set $\Lambda_S = k$, $\Lambda_G = \text{GM}_W[k]$; compute $\xi = \ln(\Lambda_S/\Lambda_G)$ and $S = \mathbf{1}[n_z \neq n_p]$; define ε via ξ , **sct** and S .

Datasets and scenarios

Experiments target: lane-keeping with crosswind; lane-change with randomly generated obstacle fields; SE(2) navigation with a car-like platform. Each scenario provides (i) nominal dynamics; (ii) disturbance models; (iii) safety sets (lane boundaries, collision sets); (iv) symmetry groups for regularization. Synthetic datasets include randomized curvature profiles, gust models and friction fields. Real-world logs (if available) can be segmented into windows W for local LTI identification.

Implementation details

Discretization via first-order hold; SQP with partial condensing for the MPC subproblems; warm-start from shifted plan; trust-region constraints on control increments. Barrier function $h(x) = p^\top x + q$ chosen per scenario; $\delta(\varepsilon)$ scheduled according to (11) with conservative constants. $\lambda_{\text{sym}}(\xi)$ linear in $|\xi|$ with saturation at $\bar{\xi}$; $N(\varepsilon)$ affine in ε within $[N_{\min}, N_{\max}]$. Composer $\mathcal{Z}$ activates curvature-speed envelopes, soft no-go sets and comfort cones conditioned on (ξ, S) .

Metrics and statistical testing

Safety (violation rate, min barrier); control (RMS tracking, jerk, effort); adaptation (re-solve time, warm-start success, average $\delta(\varepsilon)$); structure (equivariance error). For each scenario, report median and interquartile ranges over 30 randomized seeds; use Wilcoxon signed-rank tests for paired comparisons with Holm–Bonferroni correction.

Complexity and real-time performance

For $m \leq 4$ inputs, $N \in [10, 25]$, and modest state dimension, partial condensing with sparse QP solvers meets 10–20 ms budgets at 50–100 Hz control rates on embedded CPUs. The horizon/margin schedules improve worst-case latency while preserving feasibility.

Discussion

Scope and assumptions

The approach assumes local Lipschitzness and bounded identification errors over the relevant domain. Terminal sets depend on schedule bounds to ensure recursive feasibility. Discrete parity S should use hysteresis to avoid chattering at thresholds.

Broader impact and limitations

The framework prioritizes safety and feasibility; performance may be conservative when ε is large. Learning-based encoders for h and $\mathcal{Z}$ require careful validation to avoid specification gaming. Multi-agent interactions may require compositional barriers and joint invariant sets.

Future directions

Learned barrier certificates and data-driven invariant sets; multi-agent constraint composers; tighter links between ξ collapse and invariant-set growth; certified real-time implementations on embedded hardware; system identification with symmetry priors. On the sensing side, extend phase-first Doppler to *inhomogeneous* flows via path-integrated convective phase and to wideband chirps with group-delay estimators consistent with (2).

Data availability

Data supporting the findings are available from the corresponding author upon reasonable request.

Code availability

Prototype code is available from the corresponding author upon reasonable request.

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Author contributions

M.D. developed the theory, implemented the algorithms, performed analyses and wrote the manuscript.

Competing interests

The author declares no competing interests.

Materials & correspondence

Correspondence and requests for materials should be addressed to M.D.

Appendix A: Clarifying inequality directions

The discrete-time barrier recursion used throughout is

$$h(x_{t+1}) \geq (1 - \eta) h(x_t) - \delta(\varepsilon_t), \quad 0 \leq \eta < 1. \quad (12)$$

The left-hand side (next-step barrier) must be *greater than or equal to* the right-hand side to ensure forward invariance. Reversing the inequality invalidates the guarantee.

Appendix B: Symbol table

Symbol	Description
x, u	State and control input
$h(x)$	Control barrier function; $\mathcal{C} = \{x : h(x) \geq 0\}$
ξ	Continuous log-deviation $\ln(\Lambda_S/\Lambda_G)$
S	Discrete structural parity $\mathbf{1}[n_z \neq n_p]$
sct	Specific computational time (conditioning proxy)
ε	Composite conditioning index
$\delta(\varepsilon)$	Robust margin (CBF/QP feasibility buffer)
$\mathcal{Z}(\xi, S)$	Constraint composer (environment/comfort gates)
$\mathcal{X}_{\text{inv}}(\varepsilon)$	Terminal invariant set (robust positively invariant)
$\lambda_{\text{sym}}(\xi)$	Symmetry-regularization gain
$N(\varepsilon)$	Horizon schedule
$\ell(x, u; \xi)$	Stage cost (scaled by ξ)
$R_{\text{sym}}(\cdot)$	Symmetry regularizer in the objective
$f(x), g(x)$	System dynamics $x^+ = f(x) + g(x)u$
$\mathcal{U}$	Admissible control set
p, q	Linear barrier parameters when $h(x) = p^\top x + q$
η	CBF contraction factor ($0 \leq \eta < 1$)
a	Safety-filter action (QP decision variable)
$\hat{\mathbf{n}}$	Unit vector along propagation direction
c, c_s	EM and acoustic phase speeds, respectively
$\mathbf{u}$	Medium flow velocity (acoustics; lab frame)
τ	Two-way light/sound time (round-trip delay)
M	Coherent turnaround ratio
$\phi(t)$	Phase along a ray
γ	Propagation path (ray) used in the line integral
$\mathbf{k}$	Wave vector in the phase integral
$d\mathbf{x}$	Differential path element

ω	Angular frequency
$f_r(t)$	Received (instantaneous) frequency
f_0	Station clock (local oscillator)
$\Delta\phi(t)$	Two-way phase difference $\phi_\downarrow(t) - M\phi_\uparrow(t - \tau)$
$\phi_\downarrow(t), \phi_\uparrow(t)$	Downlink and (time-shifted) uplink phases
$s_\downarrow(t), s_\uparrow(t)$	Downlink and uplink carrier signals; $s_\downarrow(t) = \cos(\phi_\downarrow(t))$, $s_\uparrow(t) = \cos(2\pi f_0 t)$
$y(t)$	Fractional Doppler $y(t) = \frac{1}{2\pi M f_0} \frac{d}{dt} \Delta\phi(t)$
v_ϕ	Characteristic (phase) speed (c for EM; c_s for acoustics)
$R, \dot{R}$	Range and range rate
v_r, v_s	Receiver and source speeds along $\hat{\mathbf{n}}$ (sign per text)
W	Rolling data window for identification
$H(s)$	Identified transfer function $k \prod(s - z_i) / \prod(s - p_j)$
z_i, p_j	Zeros and poles of $H(s)$
k	Gain of $H(s)$ (note: distinct from wave vector $\mathbf{k}$)
Λ_S, Λ_G	Source and reference gains used in ξ
$G_{\text{MW}}[\cdot]$	Provenance operator (moving-window gain map)
$\mathbf{1}[\cdot]$	Indicator function
$\top$	Transpose
$\ln(\cdot)$	Natural logarithm
π	Circle constant (≈ 3.14159)
$\frac{d}{dt}$	Time derivative (dot in text denotes the same where used)

Appendix C: Phase algebra and derivations

From phase difference to fractional Doppler

Let $s_\downarrow(t) = \cos(\phi_\downarrow(t))$ and $s_\uparrow(t) = \cos(2\pi f_0 t)$, and assume coherent turnaround M . After mixing and low-pass filtering,

$$\Delta\phi(t) = \phi_\downarrow(t) - M\phi_\uparrow(t - \tau(t)). \quad (13)$$

Differentiating and normalizing by $2\pi M f_0$ yields (2). The time-variation of $\tau(t)$ carries the range-rate information, and all measurements are referenced to the single station clock.

EM two-way approximation

For small $|v_r|/c$, the round-trip delay is $\tau(t) = 2R(t)/c$ and

$$\frac{d}{dt} \Delta\phi \approx -2\pi(M f_0) \frac{2\dot{R}}{c} + \text{higher-order corrections}, \quad (14)$$

giving (3).

Acoustics with flow

With dispersion (4), the lab-frame instantaneous frequency receives $\mathbf{k} \cdot \mathbf{u}$; linearizing the two-way mapping for small speeds gives (6). For inhomogeneous $\mathbf{u}(\mathbf{x}, t)$, the convective contribution is path-integrated via $\int_{\gamma} \mathbf{k} \cdot \mathbf{u} ds$.

Appendix D: Parameter schedules

Default schedules: $\lambda_{\text{sym}}(\xi) = \min(\lambda_{\text{max}}, \lambda_0 + c_{\lambda}|\xi|)$; $N(\varepsilon) = \text{clip}(N_0 + \kappa_N \varepsilon, N_{\text{min}}, N_{\text{max}})$; trust-region radius $\Delta u(\varepsilon) = \max(\Delta u_{\text{min}}, \Delta u_0 - c_u \varepsilon)$. Hysteresis thresholds for S : $0 < \tau_{\downarrow} < \tau_{\uparrow}$.

Appendix E: Proof details for Proposition 1

Let $\hat{x}_{t+1} = f(x_t) + g(x_t)a_t^*$ be the model prediction. By Lipschitzness of h and f, g , and encoder bounds, one obtains

$$|h(x_{t+1}) - h(\hat{x}_{t+1})| \leq \epsilon_f \|p\|_1 + \epsilon_g \|p\|_1 \|a_t^*\|_1 + \epsilon_z (L_f + L_{f_{\theta}}) + \epsilon_z \|a_t^*\|_1 (L_g + L_{g_{\theta}}). \quad (15)$$

The QP constraint enforces $p^{\top} f(x_t) + p^{\top} g(x_t)a_t^* + q \geq (1 - \eta)h(x_t) - \delta(\varepsilon)$. Choosing $\delta(\varepsilon)$ as in (11) yields $h(x_{t+1}) \geq (1 - \eta)h(x_t)$.

References (narrative placeholders)

Narrative attributions appear in Background (e.g., invariant-set lane-change planners; safe-set MPC and receding-constraint MPC; tube/robust MPC and contraction metrics; symmetry via moving frames and equivariant policies; physics-driven anomaly monitors; coherent two-way Doppler in radio/sonar). Venue-ready submission should replace these with full bibliographic entries.